# Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Solving One-Variable Linear Inequalities

Solve each linear inequality for $x$ and express the solution set in interval notation:

**a** $7 - 4(2x - 3) \geq 3x - 14$ $19 - 8x \geq 3x - 14 \Rightarrow 33 \geq 11x \Rightarrow x \leq 3$, or $(-\infty, 3]$

**b** $-9 < \frac{5 - 3x}{2} \leq 6$ $-18 < 5 - 3x \leq 12 \Rightarrow -23 < -3x \leq 7 \Rightarrow -\frac{7}{3} \leq x < \frac{23}{3}$

Reverse the inequality symbol whenever multiplying or dividing by a negative number.

**2** [Foundational] • [FRQ] • Target: Compound Inequalities & Number Line Graphs

Solve the compound inequality $-5 \leq 2x - 3 < 7$ and verify that the number line below accurately represents its solution set.

Add $3$ across all three parts: $-2 \leq 2x < 10$. Divide by $2$: $-1 \leq x < 5$, which is the half-open interval $[-1, 5)$ with a closed dot at $-1$ and an open circle at $5$.

**3** [Foundational] • [FRQ] • Target: Solving Systems by Substitution

Solve the linear system by substitution and state the value of $x + y$: $y = 3x - 7$ and $4x + 3y = 31$.

Substitute $y = 3x - 7$ into the second equation: $4x + 3(3x - 7) = 31 \Rightarrow 13x - 21 = 31 \Rightarrow 13x = 52 \Rightarrow x = 4$. Then $y = 3(4) - 7 = 5$, so $(x, y) = (4, 5)$ and $x + y = 9$.

**4** [Foundational] • [FRQ] • Target: Solving Systems by Elimination

Solve the linear system by elimination: $5x - 2y = 26$ and $3x + 4y = 10$.

Multiply the first equation by $2$: $10x - 4y = 52$. Add to the second equation: $13x = 62 \Rightarrow x = \frac{62}{13}$, and $2y = 5\left(\frac{62}{13}\right) - 26 = \frac{310 - 338}{13} = -\frac{28}{13} \Rightarrow y = -\frac{14}{13}$.

**5** [Foundational] • [FRQ] • Target: Graphical Intersection of Linear Systems

The graphs of $y = 2x - 3$ and $y = -\frac{1}{2}x + 2$ are shown in the $xy$-plane. Find the exact coordinates of their intersection point $P$ and compute the area of the region bounded by the two lines and the $y$-axis.

Set $2x - 3 = -\frac{1}{2}x + 2 \Rightarrow \frac{5}{2}x = 5 \Rightarrow x = 2, y = 1$, so $P = (2, 1)$. The $y$-intercepts are $(0, -3)$ and $(0, 2)$, giving a vertical base of length $2 - (-3) = 5$ along the $y$-axis and horizontal altitude $x = 2$. Area $= \frac{1}{2}(5)(2) = 5$.

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**6** [Foundational] • [FRQ] • Target: Two-Variable System Word Problems

A university lecture hall has $320$ seats divided between standard seats ($25$ dollars each) and VIP balcony seats ($45$ dollars each). If a sold-out performance generates $9{,}600$ dollars in ticket revenue, how many seats of each type are in the hall?

Let $s$ be standard seats and $v$ be VIP seats: $s + v = 320$ and $25s + 45v = 9600$. Substitute $s = 320 - v$: $25(320 - v) + 45v = 9600 \Rightarrow 8000 + 20v = 9600 \Rightarrow 20v = 1600 \Rightarrow v = 80$ VIP seats and $s = 240$ standard seats.

**7** [Foundational] • [FRQ] • Target: Verifying Solutions to Systems of Inequalities

Determine whether the point $(3, -2)$ lies in the solution region of the system of inequalities $2x - 3y > 10$ and $x + 4y \leq -5$. Justify algebraically.

Test $(3, -2)$ in the first inequality: $2(3) - 3(-2) = 6 + 6 = 12 > 10$ (True). Test in the second inequality: $3 + 4(-2) = 3 - 8 = -5 \leq -5$ (True). Because $(3, -2)$ satisfies both inequalities simultaneously, it lies in the solution region.

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## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Dependent Systems (Infinitely Many Solutions)

Find the values of the constants $a$ and $b$ such that the linear system $ax + 6y = 15$ and $8x - 12y = b$ has infinitely many solutions.

For infinitely many solutions, the two equations must be proportional: $\frac{a}{8} = \frac{6}{-12} = \frac{15}{b} = -\frac{1}{2}$. Therefore $a = 8\left(-\frac{1}{2}\right) = -4$ and $b = 15(-2) = -30$.

**9** [Intermediate] • [FRQ] • Target: Inconsistent Systems (No Solution)

For what value of $k$ does the linear system $(k - 2)x + 4y = 9$ and $3x + (k + 2)y = 5$ have no solution?

Match ratios of coefficients: $\frac{k-2}{3} = \frac{4}{k+2} \Rightarrow k^2 - 4 = 12 \Rightarrow k^2 = 16 \Rightarrow k = 4$ or $k = -4$. At $k = 4$, $\frac{4-2}{3} = \frac{2}{3} \neq \frac{9}{5}$; at $k = -4$, $\frac{-6}{3} = -2 \neq \frac{9}{5}$. Thus both $k = 4$ and $k = -4$ yield parallel distinct lines (no solution).

**10** [Intermediate] • [FRQ] • Target: Absolute Value Inequalities & Integer Solutions

Solve the absolute value inequality $|3x - 7| + 4 \leq 15$, express the solution set as an interval, and list all integer solutions.

Subtract $4$: $|3x - 7| \leq 11 \Rightarrow -11 \leq 3x - 7 \leq 11 \Rightarrow -4 \leq 3x \leq 18 \Rightarrow -\frac{4}{3} \leq x \leq 6$. The interval is $\left[-\frac{4}{3}, 6\right]$, and the integer solutions are $\{-1, 0, 1, 2, 3, 4, 5, 6\}$ ($8$ integers).

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**11** [Intermediate] • [FRQ] • Target: Graphing & Area of Feasible Regions

The shaded region in the $xy$-plane represents the feasible set of the system $y \geq 0$, $y \leq x + 2$, and $y \leq -2x + 8$. Find the coordinates of all three corner points (vertices) of this region and the area of the feasible region.

Intersections with $y = 0$: $x + 2 = 0 \Rightarrow (-2, 0)$ and $-2x + 8 = 0 \Rightarrow (4, 0)$. Intersection of the two oblique lines: $x + 2 = -2x + 8 \Rightarrow 3x = 6 \Rightarrow (2, 4)$. The vertices are $(-2, 0)$, $(4, 0)$, $(2, 4)$, and the area is $\frac{1}{2}(4 - (-2))(4) = 12$.

**12** [Intermediate] • [FRQ] • Target: Linear Programming Optimization in Context

A coffee roaster blends Ethiopian beans ($14$ dollars/kg) and Colombian beans ($10$ dollars/kg) to make a batch of at least $40$ kg whose total ingredient cost is at most $480$ dollars. Write the system of inequalities for $(e, c)$ and find the maximum possible kilograms of Ethiopian beans in the batch.

The system is $e + c \geq 40$, $14e + 10c \leq 480$, $e \geq 0$, $c \geq 0$. To maximize $e$, make the batch size as small as allowed ($c = 40 - e$): $14e + 10(40 - e) \leq 480 \Rightarrow 4e + 400 \leq 480 \Rightarrow 4e \leq 80 \Rightarrow e \leq 20$ kg (with $c = 20$ kg).

**13** [Intermediate] • [FRQ] • Target: Structural Linear Combination Shortcut

Without solving for $x$ and $y$ individually, find the value of $x - y$ given the system $7x + 3y = 41$ and $5x + 5y = 35$.

Divide the second equation by $5$ to get $x + y = 7$. Subtract $5(x + y) = 35$ from $7x + 3y = 41$: $2x - 2y = 6 \Rightarrow x - y = 3$. (Indeed, $x = 5$, $y = 2$.)

**14** [Intermediate] • [FRQ] • Target: Relative Velocity Linear Systems

A motorboat travels $48$ km downstream in $1.5$ hours and returns the same $48$ km upstream in $2.4$ hours. Find the speed $b$ of the boat in still water and the speed $c$ of the river current.

Downstream rate: $b + c = \frac{48}{1.5} = 32$ km/h. Upstream rate: $b - c = \frac{48}{2.4} = 20$ km/h. Adding gives $2b = 52 \Rightarrow b = 26$ km/h, and subtracting gives $2c = 12 \Rightarrow c = 6$ km/h.

## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • [FRQ] • Target: Corner-Point Theorem & Linear Programming • Cross-Lesson: Topic 8.01 (Polygonal Regions & Coordinate Geometry)

Maximize the objective function $P = 5x + 8y$ subject to the linear constraints $x \geq 0$, $y \geq 0$, $x + 2y \leq 14$, and $3x + 2y \leq 26$. State the optimal point $(x^*, y^*)$ and the maximum value $P_{\max}$.

Find the corner points of the feasible polygon: $(0, 0)$, $(0, 7)$, $\left(\frac{26}{3}, 0\right)$, and the intersection of $x + 2y = 14$ and $3x + 2y = 26$, which gives $2x = 12 \Rightarrow x = 6$, $y = 4$. Evaluate $P$: $P(0, 7) = 56$, $P(6, 4) = 5(6) + 8(4) = 62$, $P(26/3, 0) = 43.\overline{3}$. Thus $P_{\max} = 62$ at $(6, 4)$.

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**16** [Challenging] • [FRQ] • Target: Linear-Quadratic System & Chord Length • Cross-Lesson: Topic 3.01 (Quadratics) & Topic 7.01 (Distance Formula)

Find all points $(x, y)$ where the line $2x - y = 1$ intersects the quadratic curve $y = x^2 - 4x + 6$, and calculate the exact length of the chord connecting the intersection points.

Substitute $y = 2x - 1$: $x^2 - 4x + 6 = 2x - 1 \Rightarrow x^2 - 6x + 7 = 0$? Let us check: if the line is $2x - y = 2 \Rightarrow y = 2x - 2$, then $x^2 - 6x + 8 = 0 \Rightarrow (x - 2)(x - 4) = 0$! With $2x - y = 1$, $x = 3 \pm \sqrt{2}$, $y = 5 \pm 2\sqrt{2}$, so $\Delta x = 2\sqrt{2}$, $\Delta y = 4\sqrt{2}$, and chord length is $\sqrt{(2\sqrt{2})^2 + (4\sqrt{2})^2} = \sqrt{8 + 32} = \sqrt{40} = 2\sqrt{10}$.

**17** [Challenging] • [FRQ] • Target: Piecewise Absolute-Value System Multiplicity • Cross-Lesson: Topic 1.01 (Absolute Value Functions)

For what values of $c$ does the system $y = |x - 2| + 1$ and $y = c$ have (a) zero solutions, (b) exactly one solution, and (c) two distinct solutions?

Since $|x - 2| \geq 0$, the minimum value of $y = |x - 2| + 1$ is $1$ at the vertex $(2, 1)$. Therefore: (a) zero solutions for $c < 1$; (b) one solution for $c = 1$; (c) two distinct solutions for $c > 1$.

**18** [Challenging] • [FRQ] • Target: Symmetric Three-Variable Linear Systems • Cross-Lesson: Topic 10.01 (Algebraic Structure & Symmetry)

Solve the three-variable linear system $x + y = 7$, $y + z = 11$, and $z + x = 10$, and compute the product $xyz$.

Add all three equations: $2(x + y + z) = 28 \Rightarrow x + y + z = 14$. Subtract each pairwise equation: $z = 14 - 7 = 7$, $x = 14 - 11 = 3$, and $y = 14 - 10 = 4$. Thus $(x, y, z) = (3, 4, 7)$ and $xyz = 3 \cdot 4 \cdot 7 = 84$.

**19** [Challenging] • [FRQ] • Target: Inscribed Circle of a Linear Feasibility Triangle • Cross-Lesson: Topic 9.01 (Circles & Inradius Formula)

The system of inequalities $x \geq 1$, $y \geq 2$, and $3x + 4y \leq 29$ defines a right triangle in the $xy$-plane. Find the coordinates of its vertices and the radius $r$ of the circle inscribed in this right triangle.

Set $x = 1$ in $3x + 4y = 29 \Rightarrow 4y = 26 \Rightarrow y = 6.5$; set $y = 2 \Rightarrow 3x = 21 \Rightarrow x = 7$. Actually, if $3x + 4y \leq 35$, then at $x = 1$, $y = 8$ (leg $\Delta y = 6$) and at $y = 2$, $x = 9$ (leg $\Delta x = 8$). With $3x + 4y \leq 29$, legs are $\Delta x = 6$ and $\Delta y = 4.5$, hypotenuse $c = \sqrt{6^2 + 4.5^2} = 7.5$, and inradius $r = \frac{a + b - c}{2} = \frac{6 + 4.5 - 7.5}{2} = 1.5$.

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**20** [Challenging] • [FRQ] • Target: Two-Component Percentage Composition System • Cross-Lesson: Topic 5.01 (Percentages & Proportions)

A nutritionist prepares a supplement mix from Powder A (30% protein, 10% fiber) and Powder B (15% protein, 25% fiber). To get exactly 180 g of protein and 110 g of fiber, how many grams of Powder A and Powder B are required?

Set up the linear system: $0.30A + 0.15B = 180$ and $0.10A + 0.25B = 110$. Multiply the second equation by 3: $0.30A + 0.75B = 330$. Subtract the first equation: $0.60B = 150 \Rightarrow B = 250$ g. Substitute into $0.10A + 0.25(250) = 110 \Rightarrow 0.10A = 47.5 \Rightarrow A = 475$ g.

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Direct Elimination in Linear Systems

From the governing equation where if $3x + 2y = 19$ and $x - 2y = 1$, what is the value of $x$?

**a.** 3 **b.** 4

**c.** 5 ✓ **d.** 6

Add the two equations directly to eliminate $y$: $4x = 20 \Rightarrow x = 5$.

**22** [Foundational] • [MCQ] • Target: Sign Reversal in Inequalities

Which of the following is the solution set to the inequality $5 - 3x > 17$?

**a.** $x < -4$ ✓ **b.** $x > -4$

**c.** $x < -\frac{22}{3}$ **d.** $x > -\frac{22}{3}$

Subtract 5: $-3x > 12$. Divide by $-3$ and flip the inequality sign: $x < -4$.

**23** [Foundational] • [MCQ] • Target: Solving Linear Systems

If $(x, y)$ is the solution to the system $2x + 5y = 24$ and $3x + 2y = 14$, what is the value of $x + y$?

**a.** 5 **b.** 6 ✓

**c.** 7 **d.** 8

Multiply the first equation by 3 ($6x + 15y = 72$) and the second by 2 ($6x + 4y = 28$). Subtracting gives $11y = 44 \Rightarrow y = 4$, and $2x + 20 = 24 \Rightarrow x = 2$. Thus $x + y = 2 + 4 = 6$.

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**24** [Foundational] • [MCQ] • Target: Translating Verbal Constraints into Inequalities

A bakery sells croissants for $3.50$ dollars each and muffins for $2.25$ dollars each. Maya buys a total of $12$ pastries and spends at most $36.00$ dollars. If $c$ is the number of croissants, which inequality represents this situation?

**a.** $3.50c + 2.25(12 - c) \leq 36.00$ ✓

**b.** $3.50c + 2.25(12 - c) \geq 36.00$

**c.** $3.50(12 - c) + 2.25c \leq 36.00$

**d.** $3.50c + 2.25c \leq 36.00$

Since Maya buys $12$ pastries in total, the number of muffins is $12 - c$. Spending 'at most $36.00$ dollars' translates to $3.50c + 2.25(12 - c) \leq 36.00$.

**25** [Foundational] • [MCQ] • Target: Testing Points in Inequality Systems

Which point $(x, y)$ lies in the solution set of the system $y > 2x - 1$ and $y < -x + 5$?

**a.** $(0, -2)$

**b.** $(1, 2)$ ✓

**c.** $(3, 4)$

**d.** $(4, 2)$

Test $(1, 2)$: $2 > 2(1) - 1 = 1$ (True) and $2 < -1 + 5 = 4$ (True).

**26** [Foundational] • [MCQ] • Target: Collinear Systems

How many solutions $(x, y)$ does the system $4x - 6y = 10$ and $-6x + 9y = -15$ have?

**a.** Zero

**b.** Exactly one

**c.** Exactly two

**d.** **Infinitely many** ✓

Divide the first equation by $2$ to get $2x - 3y = 5$, and divide the second equation by $-3$ to get $2x - 3y = 5$. Since both equations represent the exact same line, there are infinitely many solutions.

**27** [Foundational] • [MCQ] • Target: Greatest Integer Solutions to Inequalities

What is the greatest integer $x$ that satisfies the inequality $\frac{4x + 3}{5} < 7$?

**a.** $6$

**b.** $7$ ✓

**c.** $8$

**d.** $9$

Multiply by $5$: $4x + 3 < 35 \Rightarrow 4x < 32 \Rightarrow x < 8$. The greatest integer strictly less than $8$ is $7$.

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**28** [Foundational] • [MCQ] • Target: Matching Number Lines to Absolute Value Inequalities

The number line below shows the solution set to which inequality?

**a.** $|x-1| \le 3$ ✓

**b.** $|x+1| \le 3$

**c.** $|x-3| \le 1$

**d.** $|x-1| \ge 3$

The shaded interval $[-2, 4]$ has midpoint $c = \frac{-2+4}{2} = 1$ and radius $r = \frac{4-(-2)}{2} = 3$, which corresponds to $|x-1| \le 3$.

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## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

**29** [Intermediate] • [MCQ] • Target: No-Solution Condition for Linear Systems

In the system of equations $kx - 3y = 8$ and $12x - 9y = 20$, $k$ is a constant. If the system has no solution, what is the value of $k$?

**a.** $-4$

**b.** $3$

**c.** $4$ ✓

**d.** $6$

Multiply the first equation by $3$: $3kx - 9y = 24$. Comparing with $12x - 9y = 20$, the lines are parallel and distinct when $3k = 12 \Rightarrow k = 4$.

**30** [Intermediate] • [MCQ] • Target: Symmetric Linear Combination

Given the simultaneous linear system $7x - 4y = 23$ and $3x + 8y = 5$, compute the linear combination $5x + 2y$.

**a.** 12

**b.** $14$ ✓

**c.** 16

**d.** 18

Add the two equations: $(7x - 4y) + (3x + 8y) = 23 + 5 \Rightarrow 10x + 4y = 28$. Divide both sides by $2$: $5x + 2y = 14$.

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**31** [Intermediate] • [MCQ] • Target: Two-Constraint Resource Allocation

A landscaping crew can mow a small lawn in $40$ minutes and a large lawn in $75$ minutes. The crew works at most $8$ hours ($480$ minutes) per day and must mow at least $7$ lawns in total. If the crew mows $3$ large lawns, what is the maximum number of small lawns they can mow that day?

**a.** $4$ **b.** $5$

**c.** $6$ ✓ **d.** $7$

With $3$ large lawns taking $3 \times 75 = 225$ minutes, the remaining time is $480 - 225 = 255$ minutes. Since $40s \le 255 \Rightarrow s \le 6.375$, the crew can mow at most $6$ small lawns (which also satisfies $6 + 3 = 9 \ge 7$).

**32** [Intermediate] • [MCQ] • Target: Vertex Identification in Linear Inequalities

In the $xy$-plane, the solution set of the system $y \le 3x + 4$, $y \le -x + 12$, and $y \ge 2$ is a triangular region. What is the $x$-coordinate of the vertex with the greatest $y$-coordinate?

**a.** $1.5$ **b.** $2$ ✓

**c.** $2.5$ **d.** $3$

The vertex with the greatest $y$-coordinate is the intersection of the two upper boundary lines: $3x + 4 = -x + 12 \Rightarrow 4x = 8 \Rightarrow x = 2$ (where $y = 10$).

**33** [Intermediate] • [MCQ] • Target: Counting Integer Solutions

For how many integer values of $x$ is the compound inequality $11 < 4x + 3 < 35$ satisfied?

**a.** $4$ **b.** $5$ ✓

**c.** $6$ **d.** $7$

Subtract $3$: $8 < 4x < 32 \Rightarrow 2 < x < 8$. The integers strictly between $2$ and $8$ are $3, 4, 5, 6, 7$, which is $5$ integers.

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**34** [Intermediate] • [MCQ] • Target: Intersecting Inequality Constraints

A chemistry lab needs between $300\,\text{mL}$ and $450\,\text{mL}$ of acid mixture. Reagent X contains $20\%$ acid and Reagent Y contains $50\%$ acid. If the lab uses $200\,\text{mL}$ of Reagent X and $y\,\text{mL}$ of Reagent Y, and the total pure acid must be at least $115\,\text{mL}$, what is the valid range for $y$?

**a.** $100 \le y \le 250$

**b.** $120 \le y \le 250$

**c.** $150 \le y \le 250$ ✓

**d.** $150 \le y \le 300$

Total volume constraint: $300 \le 200 + y \le 450 \Rightarrow 100 \le y \le 250$. Pure acid constraint: $0.20(200) + 0.50y \ge 115 \Rightarrow 40 + 0.50y \ge 115 \Rightarrow 0.50y \ge 75 \Rightarrow y \ge 150$. Combining both gives $150 \le y \le 250$.

**35** [Intermediate] • [MCQ] • Target: Rational-Form Linear Systems

Suppose two real numbers satisfy $\frac{x+y}{x-y} = 5$ and $x + y = 10$. Determine the product $xy$.

**a.** 16

**b.** 20

**c.** 24 ✓

**d.** 25

Since $x + y = 10$, the first equation gives $\frac{10}{x-y} = 5 \Rightarrow x - y = 2$. Adding $x + y = 10$ and $x - y = 2$ gives $2x = 12 \Rightarrow x = 6$ and $y = 4$, so $xy = 24$.

**36** [Intermediate] • [MCQ] • Target: Quadrant Analysis of Linear Inequalities

In the $xy$-plane, which quadrants contain NO points $(x, y)$ satisfying the system $y > 2x + 3$ and $y > -\frac{1}{2}x + 1$?

**a.** Quadrant III only

**b.** Quadrant IV only

**c.** **Quadrants III and IV** ✓

**d.** Quadrants II and IV

The boundary lines $y = 2x + 3$ and $y = -\frac{1}{2}x + 1$ intersect at $(-0.8, 1.4)$. Since the solution region lies above both lines (which have positive and negative slopes respectively), every point $(x, y)$ in the solution set has $y > 1.4 > 0$. Therefore, no point in the solution set can lie in Quadrant III or Quadrant IV (where $y < 0$).

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**37** [Intermediate] • [MCQ] • Target: System Word Problems for Unit Prices

The cost of $4$ notebooks and $3$ pens is $24.25$ dollars, and the cost of $2$ notebooks and $5$ pens is $18.25$ dollars. What is the cost of $1$ notebook and $1$ pen combined?

**a.** $5.75$ dollars

**b.** $6.25$ dollars

**c.** $6.50$ **dollars** ✓

**d.** $6.75$ dollars

Multiply $2n + 5p = 18.25$ by $2$ to get $4n + 10p = 36.50$. Subtract $4n + 3p = 24.25$: $7p = 12.25 \Rightarrow p = 1.75$ dollars, and $2n = 18.25 - 8.75 = 9.50 \Rightarrow n = 4.75$ dollars. Thus $n + p = 4.75 + 1.75 = 6.50$ dollars.

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## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Lattice Points in a Feasible Region • Cross-Lesson: Topic 6.01 (Combinatorics & Counting)

How many ordered pairs of integers $(x, y)$ satisfy the strict inequalities $x > 0$, $y > 0$, and $2x + 3y < 14$?

**a.** $7$

**b.** $8$

**c.** $9$

**d.** $10$ ✓

Test positive integer values of $y$: for $y = 1$, $2x < 11 \Rightarrow x \in \{1, 2, 3, 4, 5\}$ ($5$ pairs); for $y = 2$, $2x < 8 \Rightarrow x \in \{1, 2, 3\}$ ($3$ pairs); for $y = 3$, $2x < 5 \Rightarrow x \in \{1, 2\}$ ($2$ pairs); for $y \geq 4$, $2x < 2 \Rightarrow$ none. Total $= 5 + 3 + 2 = 10$ pairs.

**39** [Challenging] • [MCQ] • Target: Area & Perimeter of Absolute-Value Regions • Cross-Lesson: Topic 7.01 (Isosceles Right Triangles & Special Ratios)

In the $xy$-plane, the system of inequalities $y \geq |x - 3|$ and $y \leq 5$ encloses a triangular region. What is the perimeter of this region?

**a.** $10 + 5\sqrt{2}$

**b.** $10 + 10\sqrt{2}$ ✓

**c.** $20$

**d.** $20\sqrt{2}$

The vertex of $y = |x - 3|$ is $(3, 0)$, and it intersects $y = 5$ at $(-2, 5)$ and $(8, 5)$. The horizontal top side has length $8 - (-2) = 10$, and each equal slant side has length $\sqrt{5^2 + 5^2} = 5\sqrt{2}$. Thus the perimeter is $10 + 10\sqrt{2}$.

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**40** [Challenging] • [MCQ] • Target: Determinant Condition for Unique Solutions • Cross-Lesson: Topic 10.01 (Algebraic Structure)

If the linear system $x + ky = 5$ and $kx + 4y = 10$ has a unique solution $(x, y)$, which of the following CANNOT be the value of $k$?

**a.** $-2$ only

**b.** $2$ only

**c.** $0$

**d.** $-2$ **and** $2$ ✓

A $2 \times 2$ linear system has a unique solution if and only if its determinant is nonzero:
$1(4) - k(k) \neq 0 \Rightarrow k^2 \neq 4 \Rightarrow k \neq \pm 2$.

**41** [Challenging] • [MCQ] • Target: Geometric Probability on a Feasible Region • Cross-Lesson: Topic 6.01 (Probability) & Topic 8.01 (Area)

A point $(x, y)$ is chosen uniformly at random from the rectangle $0 \leq x \leq 6$, $0 \leq y \leq 4$. What is the probability that the point satisfies $x + y \leq 4$?

**a.** $\frac{1}{4}$

**b.** $\frac{1}{3}$ ✓

**c.** $\frac{1}{2}$

**d.** $\frac{2}{3}$

The rectangle has area $6 \times 4 = 24$. The inequality $x + y \leq 4$ inside the first quadrant cuts off a right triangle with vertices $(0, 0), (4, 0), (0, 4)$ of area $\frac{1}{2}(4)(4) = 8$. Thus the geometric probability is $\frac{8}{24} = \frac{1}{3}$.

**42** [Challenging] • [MCQ] • Target: Linear-Circle Tangency System • Cross-Lesson: Topic 9.01 (Circles & Tangent Lines)

For what value of $m > 0$ does the system $y = mx$ and $x^2 + y^2 - 10x + 16 = 0$ have exactly one real solution $(x, y)$?

**a.** $\frac{3}{4}$ ✓

**b.** $\frac{4}{3}$

**c.** $\frac{3}{5}$

**d.** $\frac{5}{4}$

The circle $(x - 5)^2 + y^2 = 9$ has center $(5, 0)$ and radius $r = 3$. A tangent line $y = mx$ from the origin forms a $3{-}4{-}5$ right triangle with the radius $3$ and distance $5$, so adjacent leg $= 4$ and $\tan\theta = m = \frac{3}{4}$.

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**43** [Challenging] • [MCQ] • Target: Linear Systems in Transformed Variables • Cross-Lesson: Topic 4.01 (Exponentials & Radicals)

If $a$ and $b$ are positive real numbers satisfying $\log_2(a) + \log_2(b) = 5$ and $3\log_2(a) - \log_2(b) = 7$, what is the value of $\frac{a}{b}$?

**a.** 1

**b.** 2 ✓

**c.** 4

**d.** 8

Let $u = \log_2(a)$ and $v = \log_2(b)$. Adding $u + v = 5$ and $3u - v = 7$ gives $4u = 12 \Rightarrow u = 3$ and $v = 2$. Thus $a = 2^3 = 8$ and $b = 2^2 = 4$, so $\frac{a}{b} = 2$.

**44** [Challenging] • [MCQ] • Target: Absolute Value Diamond & Half-Plane Area • Cross-Lesson: Topic 8.01 (Area & Coordinate Symmetry)

The region defined by $|x| + |y| \le 6$ in the $xy$-plane is a square. What is the area of the portion of this square that also satisfies $y \ge x - 2$?

**a.** 36

**b.** 48 ✓

**c.** 54

**d.** 60

The square $|x| + |y| \le 6$ has diagonals of length $12$ and total area $\frac{1}{2}(12)(12) = 72$. Its four sides lie along $x - y = \pm 6$ and $x + y = \pm 6$. Since the constraint $x - y \le 2$ is parallel to the pair of sides $x - y = -6$ and $x - y = 6$, it slices the square at $\frac{2 - (-6)}{6 - (-6)} = \frac{8}{12} = \frac{2}{3}$ of its width, giving area $\frac{2}{3}(72) = 48$.

**45** [Challenging] • [MCQ] • Target: Industrial Production Linear System • Cross-Lesson: Topic 5.01 (Rates & Proportions)

A factory produces two models of solar panels, Model X and Model Y. Model X requires $3$ hours of assembly and $2$ hours of testing; Model Y requires $4$ hours of assembly and $5$ hours of testing. If the assembly department operates for exactly $108$ hours and the testing department operates for exactly $100$ hours, how many total solar panels (Model X plus Model Y) are produced?

**a.** 28

**b.** 30

**c.** 32 ✓

**d.** 35

Solve $3x + 4y = 108$ and $2x + 5y = 100$: multiply the first by $2$ ($6x + 8y = 216$) and the second by $3$ ($6x + 15y = 300$). Subtracting gives $7y = 84 \Rightarrow y = 12$, and $3x = 108 - 48 = 60 \Rightarrow x = 20$. Total panels $= x + y = 20 + 12 = 32$.

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